

Sustained Attention Scaffolds for Dense Technical Reading

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Abstract

Readers with attention-regulation differences frequently abandon research papers before extracting their core claims. We present a reading scaffold that segments a document into single-paragraph focus units, pairs each session with an explicit comprehension goal, and enforces recovery breaks on a fixed schedule. In a within-subjects study of forty-two participants, the scaffold increased completion rates from 51 percent to 88 percent and improved recall of primary findings by six tenths of a standard deviation.

1 Introduction

Dense technical prose imposes a heavy load on working memory. The reader must hold a chain of definitions, track which claims are supported by which evidence, and simultaneously resist the pull of unrelated thoughts. For many people this load is merely tiring; for readers with ADHD it is often disqualifying, and the paper is closed within minutes of being opened.

Conventional advice such as read the abstract and then skim assumes the reader can sustain a skim without derailing. We instead ask what a reading environment should actively do when sustained attention cannot be assumed. Our answer is to externalise the work of staying oriented: the environment holds the goal in view, marks the current position, and decides when it is time to rest.

2 Related Work

Spaced repetition tools address long term retention but not the initial pass through a document. Speed reading trainers optimise words per minute at a measurable cost to comprehension of argument structure. Focus timers such as the Pomodoro technique impose a working rhythm but leave the reader to manage position and meaning entirely unaided.

Typographic weighting of the leading characters of each word has a small but replicated effect on fixation stability for some readers. We treat it as one adjustable aid among several rather than as a universal default that is forced on everyone.

3 Method

The scaffold renders extracted body text as a vertical stack of paragraph units. Exactly one unit is fully legible at any moment; the neighbouring units are visibly dimmed. Advancing to the next unit is

a single deliberate keypress, which converts passive drift into an explicit choice to continue reading.

Before a session begins the reader writes one sentence describing what they need from the paper. This sentence remains pinned to the side of the screen. A fixed timer then runs twenty five minute focus blocks separated by enforced breaks, with one longer break after every fourth block.

Intrusive thoughts are captured in a visible list and deliberately deferred until the block ends. The simple act of writing the thought down reliably reduced its self reported urgency during our pilot testing.

4 Results

The completion rate rose from fifty one to eighty eight percent. Mean recall of primary findings improved by six tenths of a standard deviation, and recall of secondary details by three tenths. Time to the first meaningful written note fell by forty four percent. Participants rated the enforced break as the single most useful component, followed closely by the one paragraph spotlight.

The optional auto pace mode, which advances units at a fixed reading speed, was adopted by only a minority of participants and produced no measurable comprehension gain, although several people reported that it felt calming during long sessions.

5 Discussion

The scaffold works by moving orientation effort out of the reader's head and into the interface itself. It does not make the underlying prose any easier; it makes losing your place harder. This framing predicts that the largest gains should appear on the longest and least well structured papers, which is exactly what we observe in the data.

A clear limitation is that extracted text discards figures, tables, and the layout of equations. For heavily mathematical papers the original page view remains necessary, and in that mode the scaffold degrades gracefully to a timer paired with a section checklist.

6 Conclusion

Treating sustained attention as a resource that the environment must supply, rather than one the reader alone must provide, yields a simple reading tool with surprisingly large effects on completion. Future work will test adaptive block lengths and direct integration with common reference managers.

References

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